

A THEORY OF ECONOMIC SLACK

Pascal Michailat

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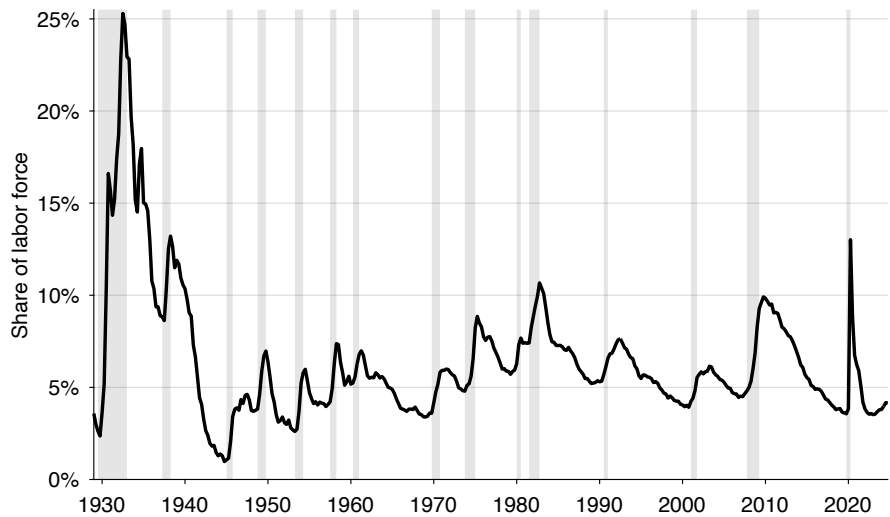
Draft manuscript available at <https://pascalmichailat.org/18/>

SLACK IS ALMOST EVERYWHERE AT ALL TIMES

JOBSEEKERS QUEUEING DURING THE GREAT DEPRESSION (DOROTHEA LANGE)



AVERAGE US UNEMPLOYMENT = 6.4% (PETROSKY-NADEAU, ZHANG 2021)



UNSOLD GOODS AND SERVICES ON THE PRODUCT MARKET



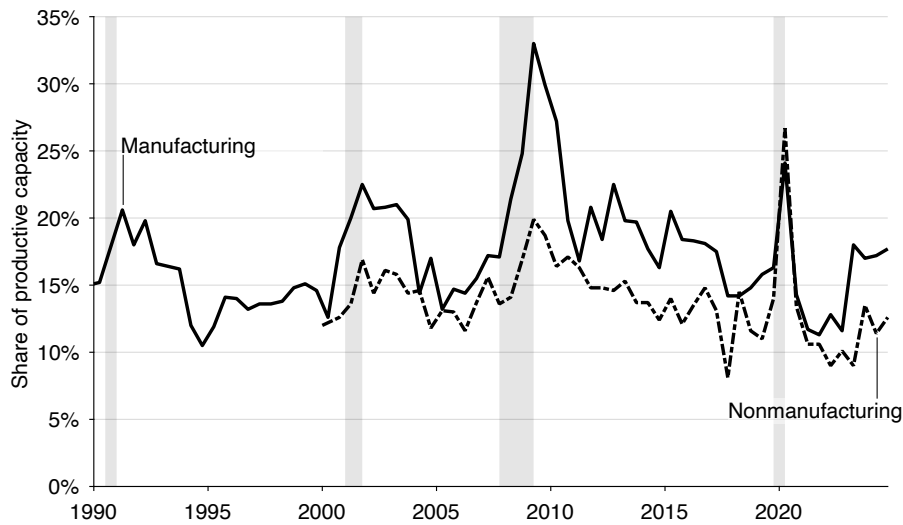
UNSOLD GOODS AND SERVICES ON THE PRODUCT MARKET



UNSOLD GOODS AND SERVICES ON THE PRODUCT MARKET



AVERAGE US IDLENESS = 14%–17% (ISM)



YET SLACK VANISHED FROM BUSINESS CYCLE MODELS

	Keynes (1936)	Gali (2008)	Woodford (2003)
“Unemployment” in index	5–9, 13, 15, 16, 22, 109, 121, 128, 190, 235, 251, 274, 275, 278, 289, 334, 335, 346–349, 358, 359, 362–364, 381, 382	188	Ø
“Inflation” in index	119, 202, 207, 281, 291, 301, 303, 331, 332	1, 4, 11, 21–22, 31–32, 34, 37–38, 47, 49–52, 54–56, 60, 77, 79, 95–99, 103, 105, 107–108, 116, 120, 125– 127, 130, 133–135, 137– 139, 148, 155–156, 161– 164, 168–176, 179, 189	3–5, 13–14, 17, 19–21, 39, 43–44, 46, 49, 90– 101, 116–122, 126–138, 159–160, 176–177, 188, 203–205, 212–213, 215, 236, 248, 251, 262, 267, 272, 276–286, 289–295, 301–305, 317, 341, 347, 360–362, 381–382, 396–405, 408–409, 413–416, 418, 437, 440–441, 445– 446, 460, 462–463, 467–485, 493, 499–501, 522–527, 544, 559–565, 576–582, 590–604, 615, 619–623, 639–641, 694–695, 714–715, 724–727
Inflation share	22%	98%	100%

OVERVIEW OF THE BOOK

FIRST HALF: INTRODUCTION + SLACKISH MARKETS

1. Introduction: slack vanished when Keynesian model → New Keynesian model
2. Data: prevalence, cyclical, + welfare cost of slack
3. Matching function: assumption + examples
- 4–7. Static slackish market: demand + supply, price norms, comparative statics
8. Dynamic slackish market: customer relationships
9. Market efficiency + inefficiency
- 10–12. Application to the labor market: job rationing, state dependence, full employment

SECOND HALF: SLACKISH BUSINESS CYCLES + CONCLUSION

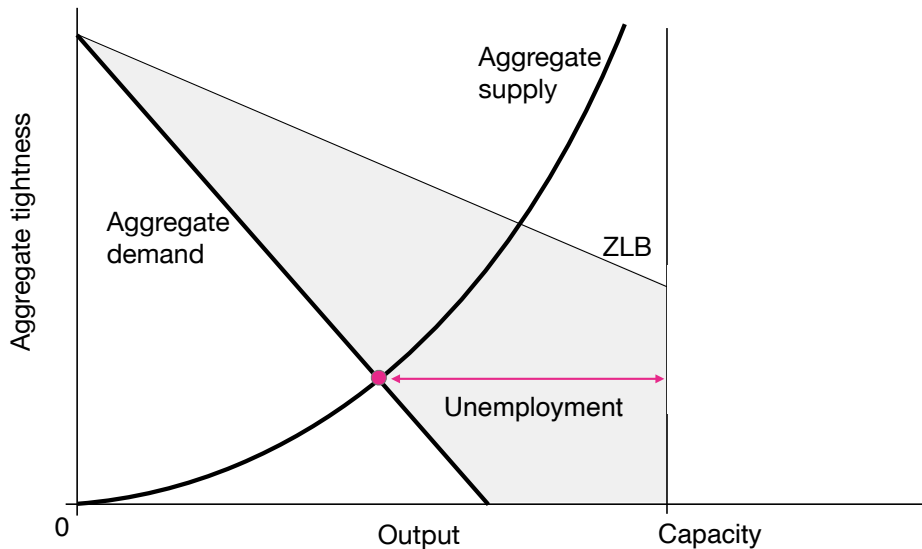
- 13. Slackish business cycle model: interest-rate policy + forward guidance
- 14. Slackish Phillips curve: convexity + divine coincidence
- 15. Slackish labor + product markets: Keynesian + classical + frictional unemployment
- 16. Optimal monetary policy: based on slack gap + monetary multiplier
- 17. Optimal fiscal policy: based on slack gap + fiscal multiplier
- 18. Recession detection using slack
- 19-20. Takeaways + open questions
- 21. Slack around the world
- 22. Kuhnian review of business cycle research

OVERVIEW OF MONETARY POLICY RESULTS

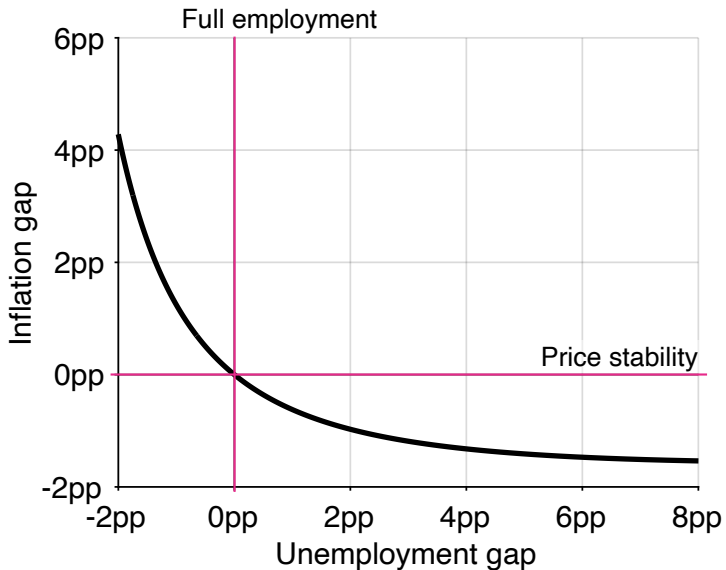
1. **No Taylor rule:** Fed can set a nominal/real interest rate
2. **Not topsy-turvy ZLB:** economy behaves normally at ZLB
3. **No forward-guidance puzzle:** forward guidance weakens as ZLB lasts longer
4. **Not marginal costs:** Phillips curve links unemployment to inflation
5. **No cost-push shocks:** shifts in Beveridge curve $\leadsto$ shifts Phillips curve
6. **No linear Phillips curve:** Phillips curve is sharply convex
7. **Divine coincidence:** full employment + price stability mandates overlap
8. **No NAIRU:** full-employment target is FERU, given by $u^* = \sqrt{uv}$ in the US
9. Divine coincidence fails if normal inflation drifts: unemployment/inflation tradeoff
10. Monetary policy suboptimal at least half of time: unemployment gap $\times$ inflation gap < 0

PHILLIPS CURVE

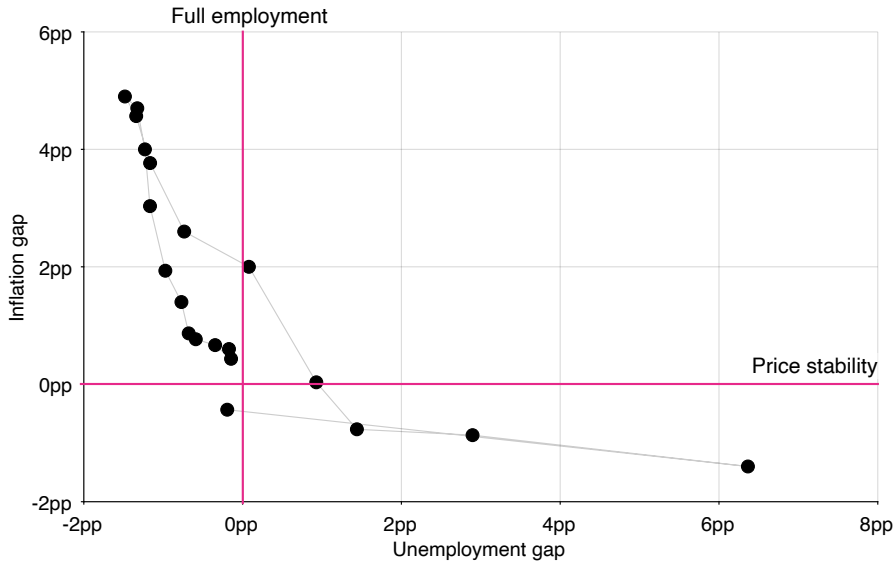
REAL RATE $\rightsquigarrow$ AGGREGATE DEMAND $\rightsquigarrow$ UNEMPLOYMENT ...



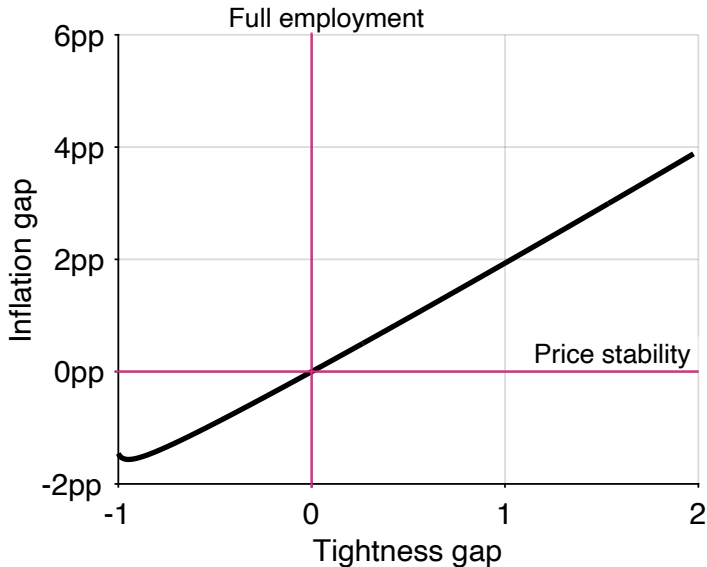
...UNEMPLOYMENT $\rightsquigarrow$ INFLATION



US PHILLIPS CURVE AFTER THE PANDEMIC

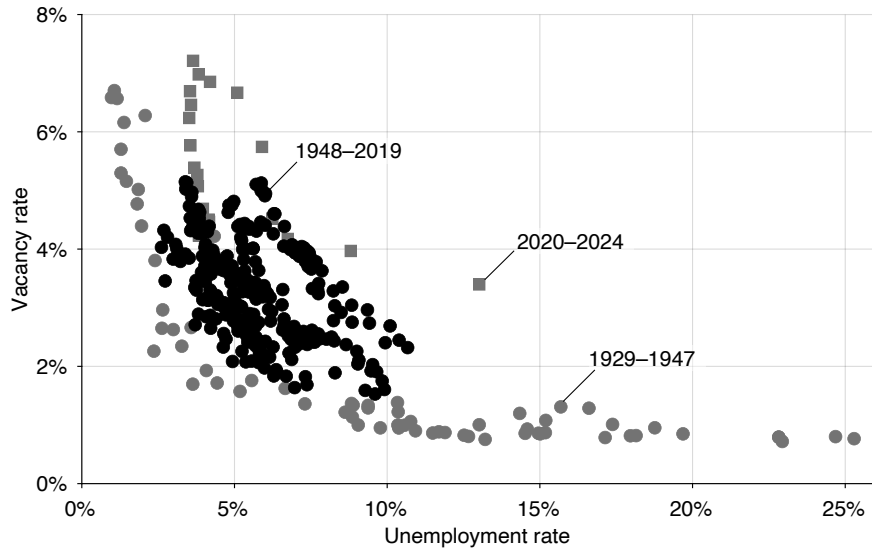


PHILLIPS CURVE IS CONVEX IN UNEMPLOYMENT BUT LINEAR IN TIGHTNESS

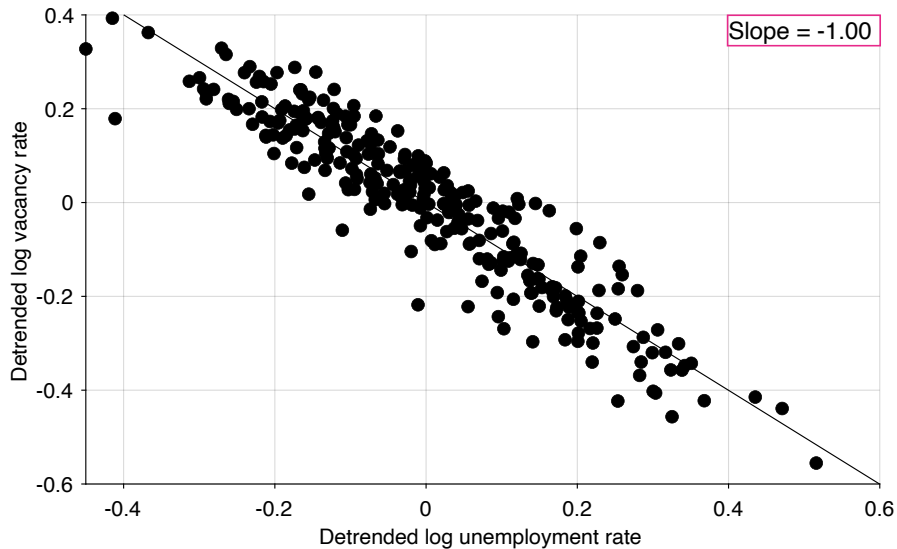


BEVERIDGE CURVE

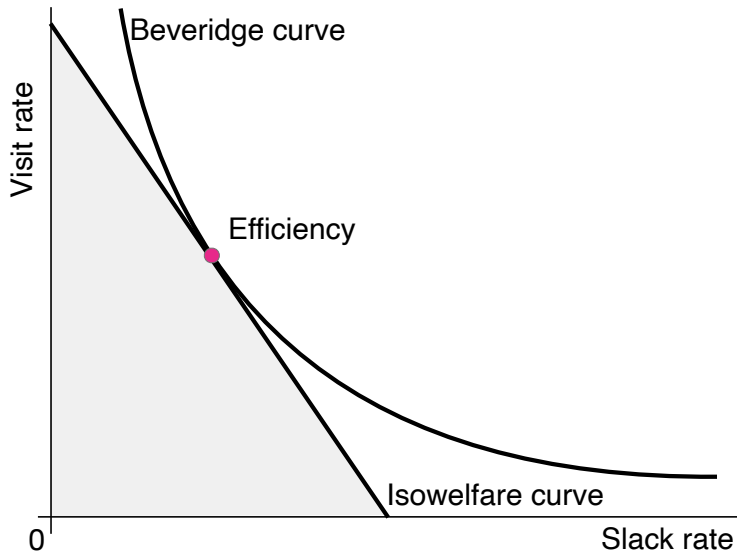
US BEVERIDGE CURVE, 1929–2024



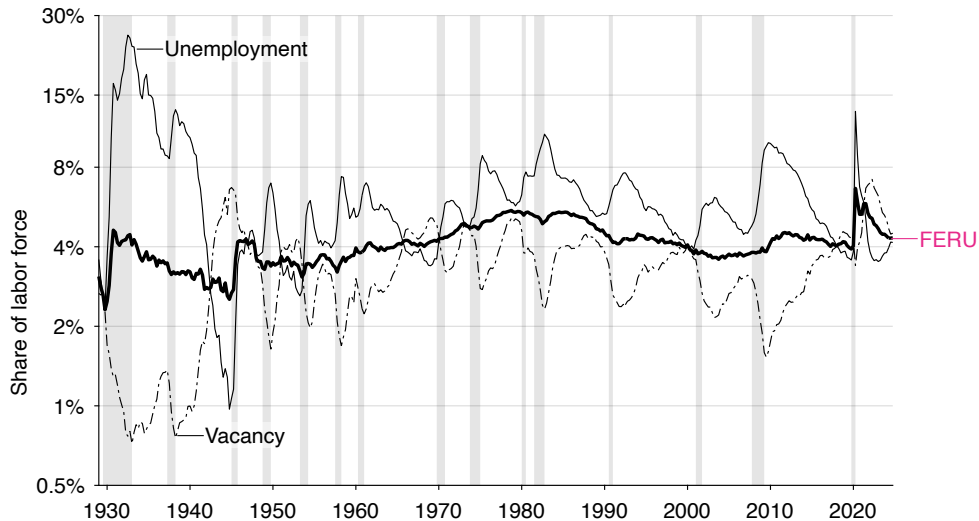
US BEVERIDGE CURVE = RECTANGULAR HYPERBOLA



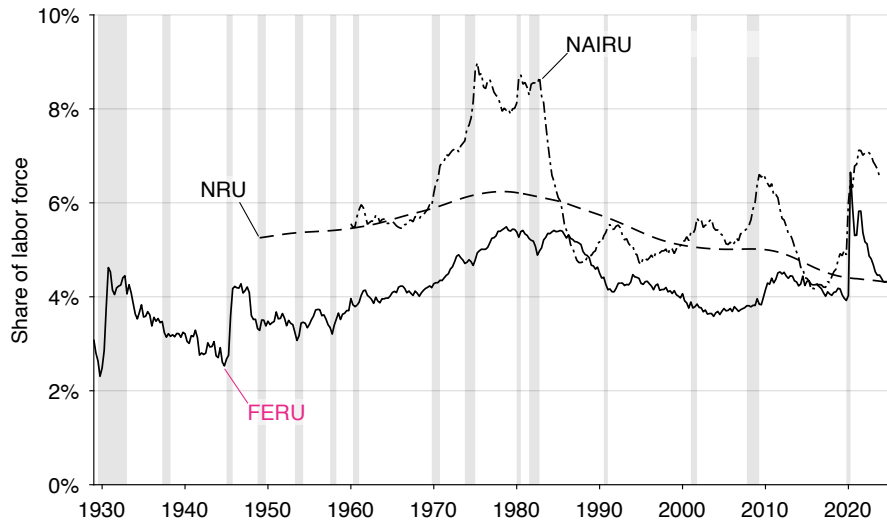
SLOPE OF BEVERIDGE CURVE DETERMINES EFFICIENT UNEMPLOYMENT RATE



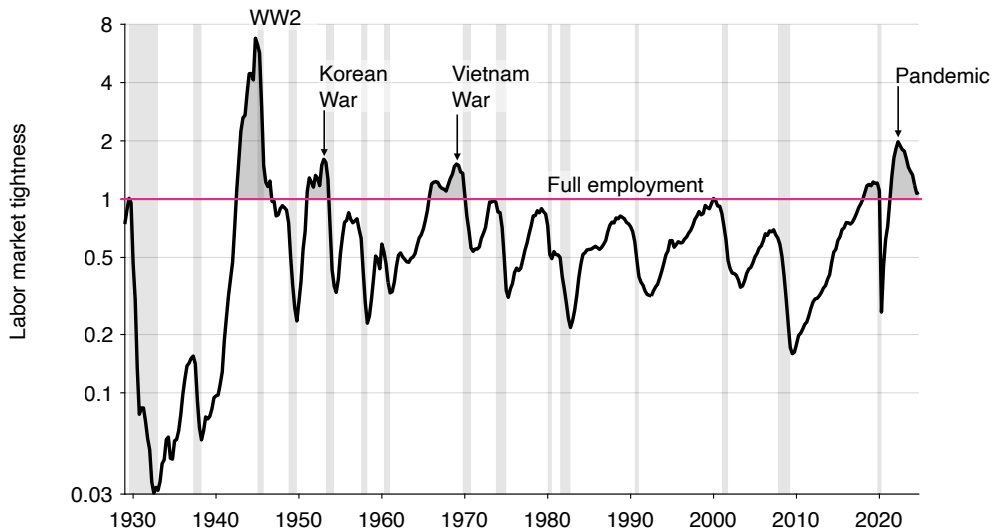
FULL-EMPLOYMENT RATE OF UNEMPLOYMENT (FERU): $u^* = \sqrt{uv}$



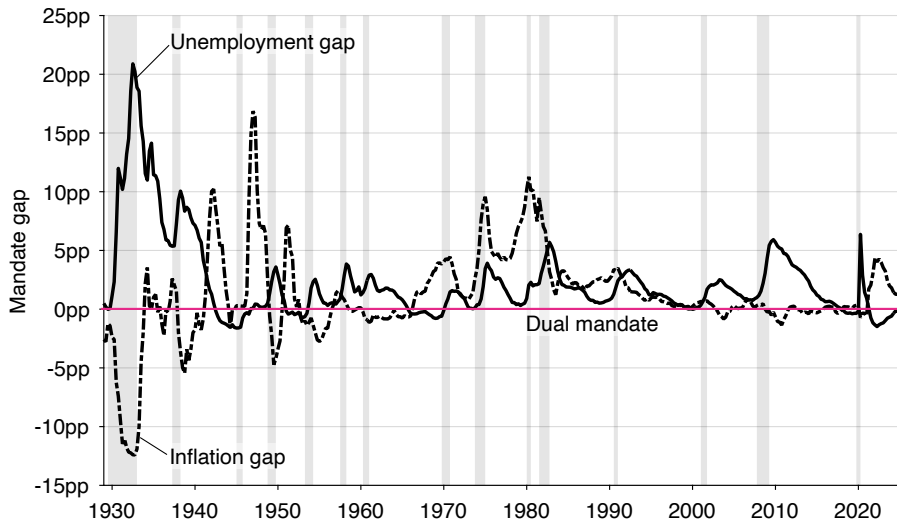
FERU AVERAGES 4.1%—BELOW OTHER UNEMPLOYMENT TARGETS



US LABOR MARKET IS GENERALLY INEFFICIENTLY SLACK: $v/u < 1$



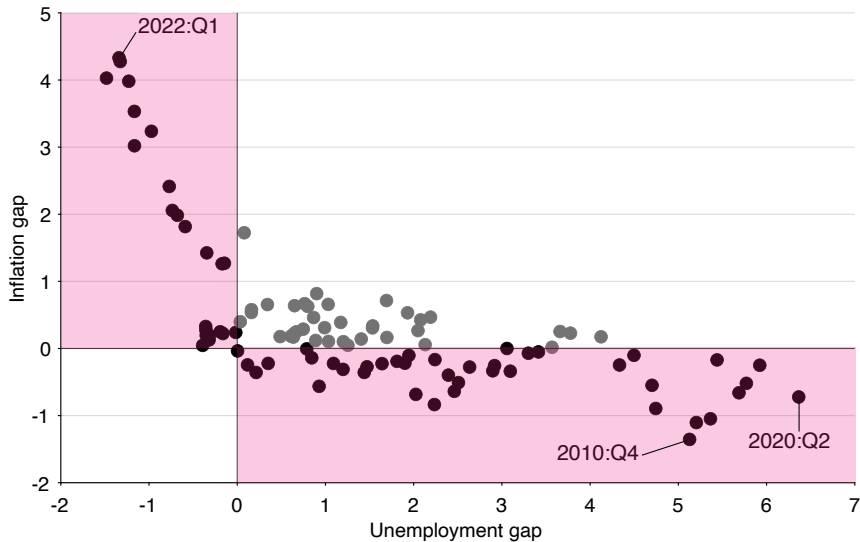
DEVIATION FROM DUAL MANDATE IN THE US, 1929–2024



1929–2024: MONETARY POLICY IS SUBOPTIMAL > 47% OF TIME

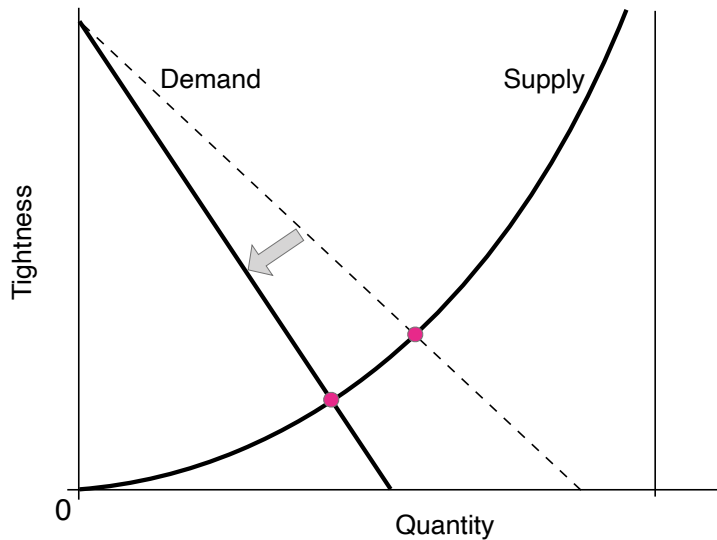


2000–2024: MONETARY POLICY IS SUBOPTIMAL > 63% OF TIME

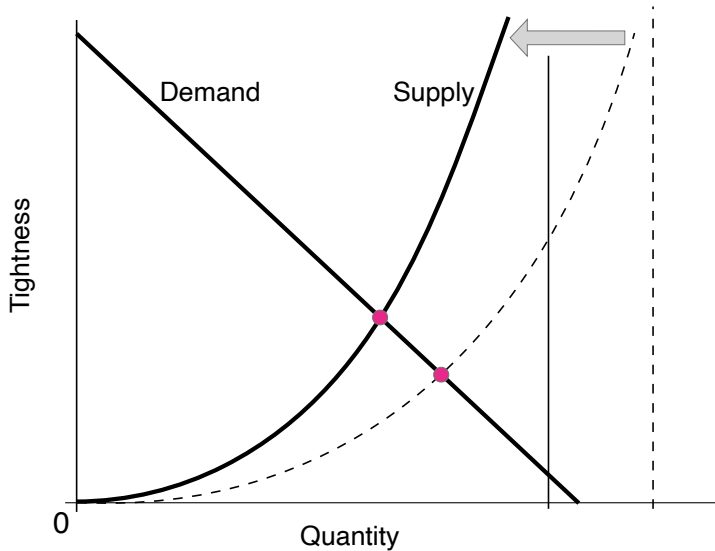


OKUN CURVES

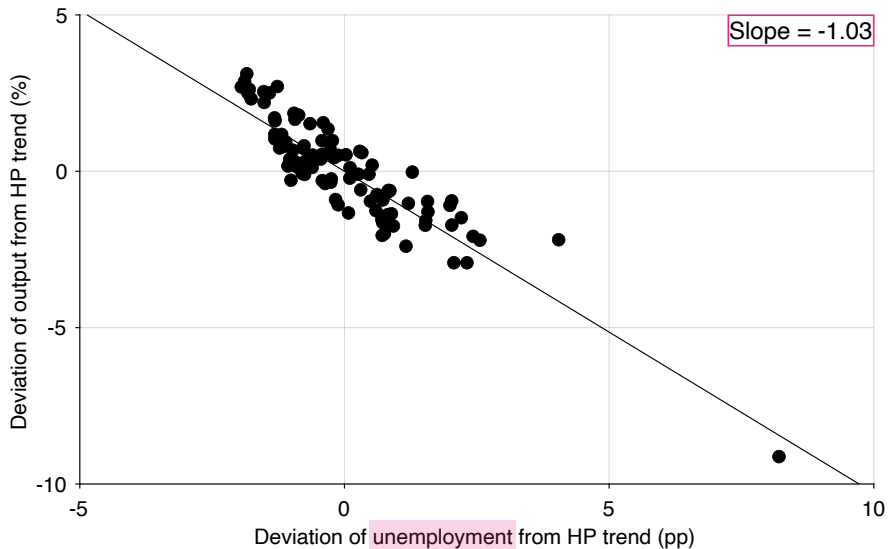
DEMAND SHOCK: NEGATIVE SLACK-OUTPUT CORRELATION



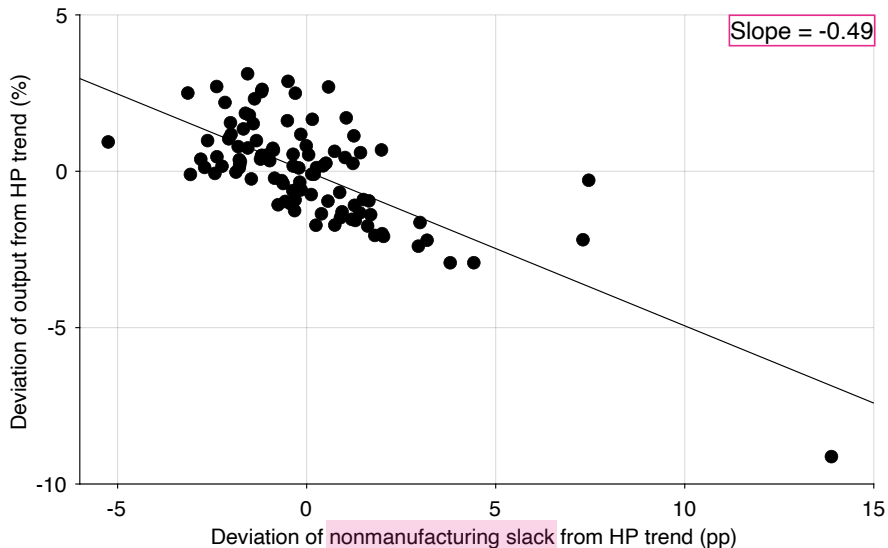
SUPPLY SHOCK: POSITIVE SLACK-OUTPUT CORRELATION



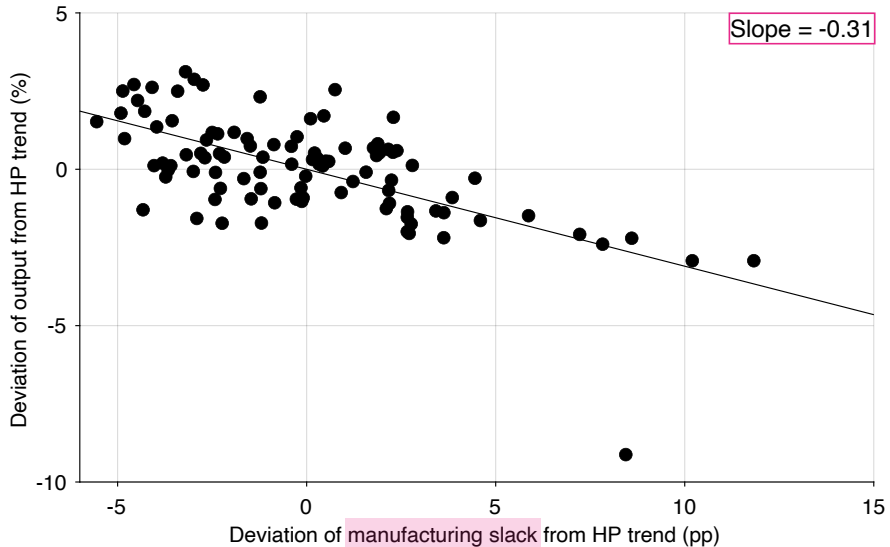
LABOR OKUN CURVE: LABOR DEMAND SHOCKS ARE PREDOMINANT



PRODUCT OKUN CURVE: AGGREGATE DEMAND SHOCKS ARE PREDOMINANT



PRODUCT OKUN CURVE: AGGREGATE DEMAND SHOCKS ARE PREDOMINANT



FISCAL POLICY RESULTS

1. **Fiscal multipliers are not constant:** sharply slack dependent
 - Demand-side policies are more effective in slack markets
 - Supply-side policies are more effective in tight markets
2. **Optimal stimulus does not depend on model details:** only 3 sufficient statistics
 - Slack gap
 - Fiscal multiplier
 - Elasticity of substitution between public and private goods
3. **Stimulus does not require multiplier > 1 :** only > 0
4. **Optimal stimulus does not completely fill the unemployment gap:** only partially
5. **Optimal stimulus is not increasing in multiplier:** hump-shaped

OPTIMAL STIMULUS SPENDING DURING GREAT DEPRESSION

